MACHINE LEARNING OPERATIONS

Assignment 01
End-to-End Heart Disease ML System

Data acquisition | *Experiments* | *CI/CD* | *Docker* | *API* | *Monitoring*

Item	Implementation
Dataset	UCI Heart Disease - processed Cleveland subset
Target	Binary: original 0 -> absence; original 1-4 -> presence
Models	Logistic Regression and Random Forest
Experiment tracking	MLflow runs, params, metrics, CV tables, plots, model artifacts
Serving / monitoring	FastAPI + Docker + Prometheus + Grafana; Kubernetes manifests
Github Code	https://github.com/dileep3004/heart-disease-mlops
Video Recording	https://drive.google.com/file/d/1iz4v1glSiNL3mhgisnIm99jh02N5JP9l/view?usp=sharing

Executive summary

This project applies a reproducible MLOps workflow to a small historical tabular dataset. A downloader extracts the official UCI file, a cleaning step normalizes missing-value markers and derives the binary target, and an EDA script generates class-balance, feature-distribution, missingness, and correlation plots. Two scikit-learn pipelines are tuned using stratified cross-validation and evaluated on a held-out test set. MLflow tracks experiments, while FastAPI, Docker, CI, Kubernetes manifests, request logs, Prometheus metrics, and a provisioned Grafana dashboard cover delivery and operational visibility.

Important: all reported performance values must come from a successful run on the downloaded dataset. This report generator inserts values from reports/metrics.json when available; it does not manufacture metrics when the external dataset has not been downloaded.

1. Problem, Dataset, and Assumptions

The assignment asks for a binary classifier that consumes patient health features and predicts whether heart disease is present, then deploys the model as a cloud-ready monitored API. This is implemented as a coursework demonstration, not a clinically validated prediction system.

Dataset source and schema

The official UCI Heart Disease dataset page describes four source databases and 303 instances in the processed Cleveland subset. It also explains that the original goal field takes values 0 to 4 and that published Cleveland experiments commonly distinguish 0 (absence) from 1-4 (presence). The implementation uses the no-header processed Cleveland file from the UCI archive and declares 13 predictor columns plus the original goal field.

Decision	Reason / consequence
Cleveland subset	Keeps to the classic 303-record processed file, suitable for this assignment's scope.
Binary target	Maps original goal 0 to 0 and any positive goal to 1.
Missing features	`?` is normalized to NaN; imputation is fit inside CV/training pipelines.
Train/test protocol	Stratified 80/20 split, random_state=42; test set is not used for tuning.
Model selection	5-fold StratifiedKFold; mean validation ROC-AUC is the selection metric.
Reproducibility	A fixed random seed, parameter grids, requirements file, and data hash manifest.

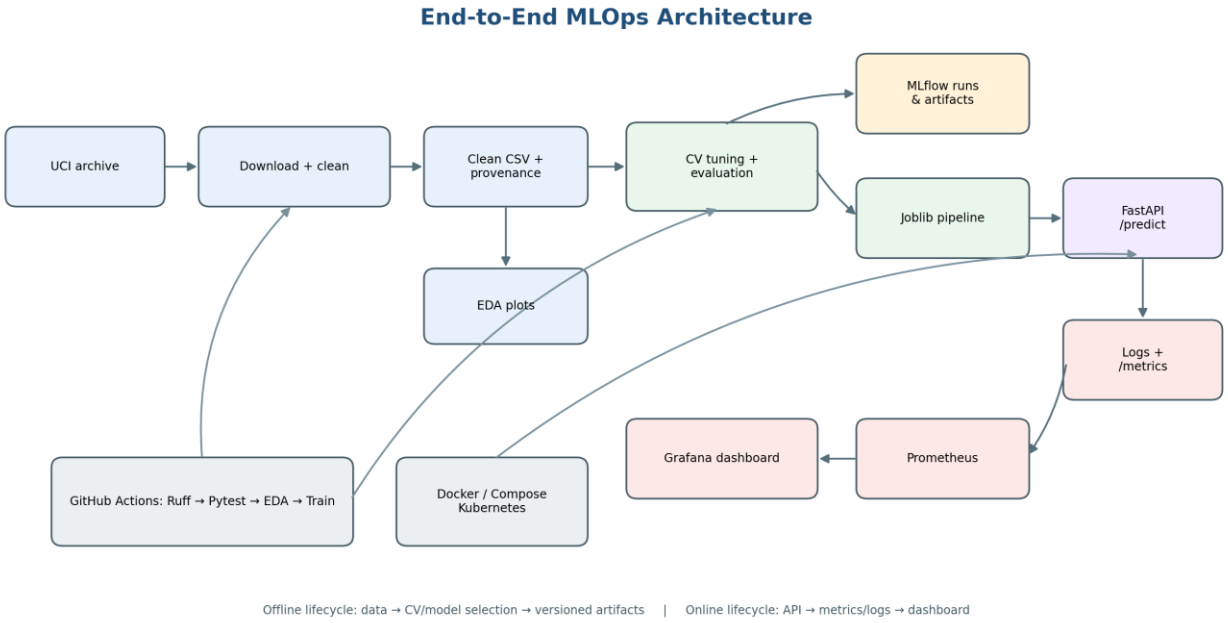
Known limitations

- Historical, small dataset; the split metrics have high sampling uncertainty and are not evidence of clinical utility.
- Feature names and coded values need domain context; input validation enforces basic ranges but is not medical validation.
- The API intentionally returns an educational model estimate and a clear not-for-diagnosis disclaimer.

2. System Architecture

Data plane, model lifecycle, and delivery path

The workflow separates offline model development from online inference. The raw archive is normalized to a stable tabular schema, and preprocessing stays attached to each estimator so the serving service can load one serializable object and apply the same transformations that were fit during training.



Principal components

Component	Responsibility
Data/EDA	Fetch, clean, record provenance, summarize missingness and distributions.
Training	Grid search for two pipelines, CV and holdout metrics, plots and model selection.
MLflow	Model-family runs with parameters, metrics, cross-validation tables, evaluation plots and model artifacts.
Serving	FastAPI schema validation, prediction, probability, health/readiness routes and safe request logs.
Operations	Docker image, Compose-based Prometheus/Grafana stack, Kubernetes Deployment/Service.
CI	Ruff, pytest, data acquisition, EDA, training, and artifact upload on GitHub Actions.

3. Data Acquisition, Cleaning, and EDA

Run `python -m heart_mlops.data --output data/processed/heart_disease_clean.csv`. The downloader retrieves the UCI ZIP archive, locates the processed Cleveland file, extracts the raw no-header CSV, writes a named cleaned dataset, and saves a JSON provenance manifest. The manifest records source URL, raw-file SHA-256, row count, feature count, missing feature cells, and class counts.

Cleaning rules

- Use an explicit 14-column schema rather than relying on column inference.
- Normalize `?` markers and invalid numeric strings to missing values.
- Drop rows with missing targets; derive `target = 1` when original `num > 0`, otherwise 0.
- Drop exact duplicate rows and persist features in a fixed order; preserve feature NaNs for train-fold imputation.

EDA outputs

Plot	Question answered
Numeric histograms by target	How do the five continuous features distribute by class?
Target balance	How many records map to absence vs presence?
Numeric correlation heatmap	What pairwise linear relationships appear among continuous measurements?
Missingness bars	Which feature fields need imputation and how many cells are missing?

Execute `python -m heart_mlops.eda --input data/processed/heart_disease_clean.csv --output reports/figures`. Numeric-only correlation is chosen to avoid presenting ordinal/coded categorical values as continuous measurements. Interpret the plot as exploratory association, not causation.

Observed values are generated in `reports/eda_summary.json`; do not copy general expected class ratios into the submission. Inspect the images after download and write two or three dataset-specific observations based on the actual output.

4. Feature Engineering and Model Development

The 13 fields are separated into five continuous/numeric measurements (age, trestbps, chol, thalach, oldpeak) and eight coded categorical fields (sex, cp, fbs, restecg, exang, slope, ca, thal). The data are not globally scaled or imputed before splitting. Instead, each model contains a ColumnTransformer fitted only on training folds.

Step	Logistic Regression	Random Forest
Numeric missing values	Median imputation	Median imputation
Categorical missing values	Most-frequent imputation	Most-frequent imputation
Categorical encoding	OneHotEncoder, unknown categories ignored	OneHotEncoder, unknown categories ignored
Numeric scaling	StandardScaler	No scaling
Hyperparameter grid	C in {0.1, 1, 10}; class_weight in {None, balanced}	n_estimators in {100, 250}; max_depth in {None, 6}; min_samples_leaf in {1, 3}; class_weight in {None, balanced}

Protocol

- Use a stratified 80/20 train/test split with random_state=42.
- Tune using 5-fold StratifiedKFold with shuffle=True and random_state=42.
- Score candidate configurations with ROC-AUC, accuracy, precision, and recall; refit the best mean ROC-AUC configuration.
- Report holdout accuracy, precision, recall, F1, and ROC-AUC plus the confusion matrix, ROC curve, classification report, and transformed feature importance.
- Select the model family by cross-validation ROC-AUC, not by searching for the best-looking holdout score.

ROC-AUC captures ranking quality across thresholds; precision describes the fraction of positive predictions that are correct; recall describes positive-class coverage. These metrics can conflict, so results should be discussed in context and not reduced to accuracy alone.

5. Experiment Results and Model Selection

Model	CV ROC-AUC mean +/- SD	Test acc.	Test prec.	Test recall	Test ROC-AUC
LogisticRegression	0.903 +/- 0.017	0.885	0.839	0.929	0.966
RandomForest	0.903 +/- 0.030	0.869	0.833	0.893	0.952

Selected family: LogisticRegression. The selection is based on the highest 5-fold mean validation ROC-AUC. The complete measured report, best parameters, classification reports, and data split metadata are in `reports/metrics.json`; per-configuration CV results are in the `artifacts/` directory and MLflow.

How to interpret the result

- Compare cross-validation mean and standard deviation to assess both average quality and fold sensitivity.
- Compare holdout precision and recall to show the practical trade-off for classifying the positive class.
- Use the confusion matrix to identify false positives and false negatives; do not infer medical harm or benefit from this small dataset alone.
- Use feature-importance plots as model diagnostics, not as causal explanations.

6. Experiment Tracking and Reproducibility

The training script creates an MLflow experiment named `heart-disease-classification`, with one parent run per model family and a nested child run for every GridSearchCV parameter configuration. Child runs log the candidate hyperparameters and the mean/standard deviation for validation ROC-AUC, accuracy, precision, and recall. Each parent run includes selected hyperparameters, split/CV settings, final holdout metrics, a fitted scikit-learn pipeline, JSON evaluation summary, classification report, plot artifacts, and the complete CV candidate-results table.

Artifact	Purpose
`models/heart_disease_model.joblib`	Serializable preprocessing + classifier pipeline used by the API.
`artifacts/*_cv_results.csv`	Per-configuration mean/std train-fold metrics and hyperparameters.
`artifacts/*_classification_report.json`	Precision, recall, F1 and support by target class.
`artifacts/model_metadata.json`	Selected model family, feature order and target mapping.
`reports/metrics.json`	Consolidated model metrics, selection protocol and limitations.
`data/processed/*.manifest.json`	Dataset provenance, source and SHA-256 digest.

Reproduce the run

Create a clean Python 3.11 virtual environment, install `requirements.txt`, install the project with `pip install -e .`, download the data, then run the EDA and training module commands in the README. `random_state=42` is used for splitting, CV shuffling, and estimator seeds where applicable. Exact metric values can still vary across library versions or numerical backends, so the installed environment and final output files should be retained.

Run `mlflow ui --backend-store-uri ./mlruns --host 127.0.0.1 --port 5000` to inspect the local tracking store. The default store is local for the demonstration; a shared remote tracking server and artifact store would be required for multi-developer production use.

7. Automated Testing, CI/CD, and Containerization

Testing and linting

Ruff checks style and common defects. Pytest covers target binarization, missing-value preservation, pipeline fitting and probability output, API health and Prometheus metrics, prediction payload validation, and successful inference using a small deterministic test model. The test suite does not rely on external dataset access, allowing quick deterministic feedback on a laptop or CI runner.

GitHub Actions flow

Stage	Behavior / failure condition
Checkout + Python 3.11	Starts with a clean runner and pip install from requirements.
Ruff	Workflow fails if selected lint rules report errors.
Pytest	Workflow fails if a unit or API test fails.
Dataset + EDA	Workflow downloads official data then creates plots/summary.
Train + tracking	Tuning, held-out evaluation and MLflow artifacts run after tests.
Upload artifacts	Stores reports, model/evaluation artifacts and processed CSV for 14 days.

Docker packaging

The Dockerfile uses Python 3.11 slim and a separate `requirements-api.txt` to avoid installing MLflow and developer tooling in the serving image. It copies the service source and the trained joblib artifact, sets `MODEL_PATH`, exposes port 8000, and includes a health check. Train the model first; the Docker build deliberately fails if the expected artifact does not exist.

Run `docker compose up --build -d`; verify the API docs at `http://localhost:8000/docs`, then check `docker compose ps` and logs. Never put credentials, private data, or a production patient record in source control or image layers.

8. Deployment, Logging, and Monitoring

API contract

Endpoint	Purpose
`GET /health`	Process liveness; does not imply a trained model is present.
`GET /ready`	Readiness check for the model artifact.
`POST /predict`	Validates all 13 fields and returns class, class-1 probability and confidence.
`GET /metrics`	Prometheus counters and request latency histogram.

Operational metrics

- Request counter labelled by method, path, and HTTP status code.
- Latency histogram supporting p95 latency calculation.
- Prediction counter split by predicted class, without recording patient features.
- Request logs contain method/path/status/elapsed milliseconds, not the input body.

Docker Compose starts Prometheus and Grafana and provisions a dashboard for request rate, p95 latency, prediction counts and HTTP 5xx rates. Generate sample traffic from Swagger or the supplied client, wait at least two scrape intervals, and verify Prometheus's target status is UP. The dashboard is operational visibility, not data/model drift monitoring; no ground-truth feedback loop exists in this static demo.

Kubernetes

`k8s/deployment.yaml` defines two API replicas, resource requests/limits, readiness and liveness probes, scrape annotations, and a LoadBalancer Service. For Minikube, load the image into the cluster and use `kubectl port-forward svc/heart-disease-api 8000:80`. A real external LoadBalancer depends on support in the target environment; a manifest alone does not prove public cloud deployment.

Production gaps

Before any non-demo deployment, add HTTPS/TLS, authentication and authorization, rate limiting, network restrictions, secret management, audit controls, privacy review, model/version registry, alerting, monitoring of input drift and observed outcomes, and governance approvals. This health-related dataset is not clinically valid for decision support.

9. Demo Runbook, Rubric Mapping, and References

A complete demonstration should be recorded from a clean environment and should show the outputs rather than only the source code. Execute the following commands from the repository root in the order shown.

Step	Command / evidence
1. Setup	`pip install -r requirements.txt` then `pip install -e .`
2. Acquire data	`python -m heart_mlops.data --output data/processed/heart_disease_clean.csv`
3. EDA	`python -m heart_mlops.eda --input data/processed/heart_disease_clean.csv --output reports/figures`
4. Train	`python -m heart_mlops.train --input data/processed/heart_disease_clean.csv --artifacts-dir artifacts --reports-dir reports --mlflow-dir mlruns`
5. Report	`python scripts/generate_report.py` (reads measured metrics when present)
6. Quality gates	`ruff check src tests scripts` and `pytest`
7. API	`uvicorn heart_mlops.api:app --reload --port 8000`; test `/docs`, `/predict`, `/metrics`
8. Monitor	`docker compose up --build -d`; capture Prometheus and Grafana screenshots
9. Deploy/CI	Push to GitHub to run Actions; optionally apply the Minikube manifest and capture `kubectl` output.

Assignment deliverable map

EDA and data download are implemented in `data.py` and `eda.py`; feature engineering, two models and CV tuning are in `modeling.py` and `train.py`; MLflow logs experiments; joblib + requirements provide a reusable model and environment; pytest/Ruff/GitHub Actions cover automated quality checks; Docker and Kubernetes files implement packaging/deployment; request logs and Prometheus/Grafana cover monitoring; this report plus architecture diagram cover documentation. Record your actual repository URL and public API URL only after creating/validating them.

References

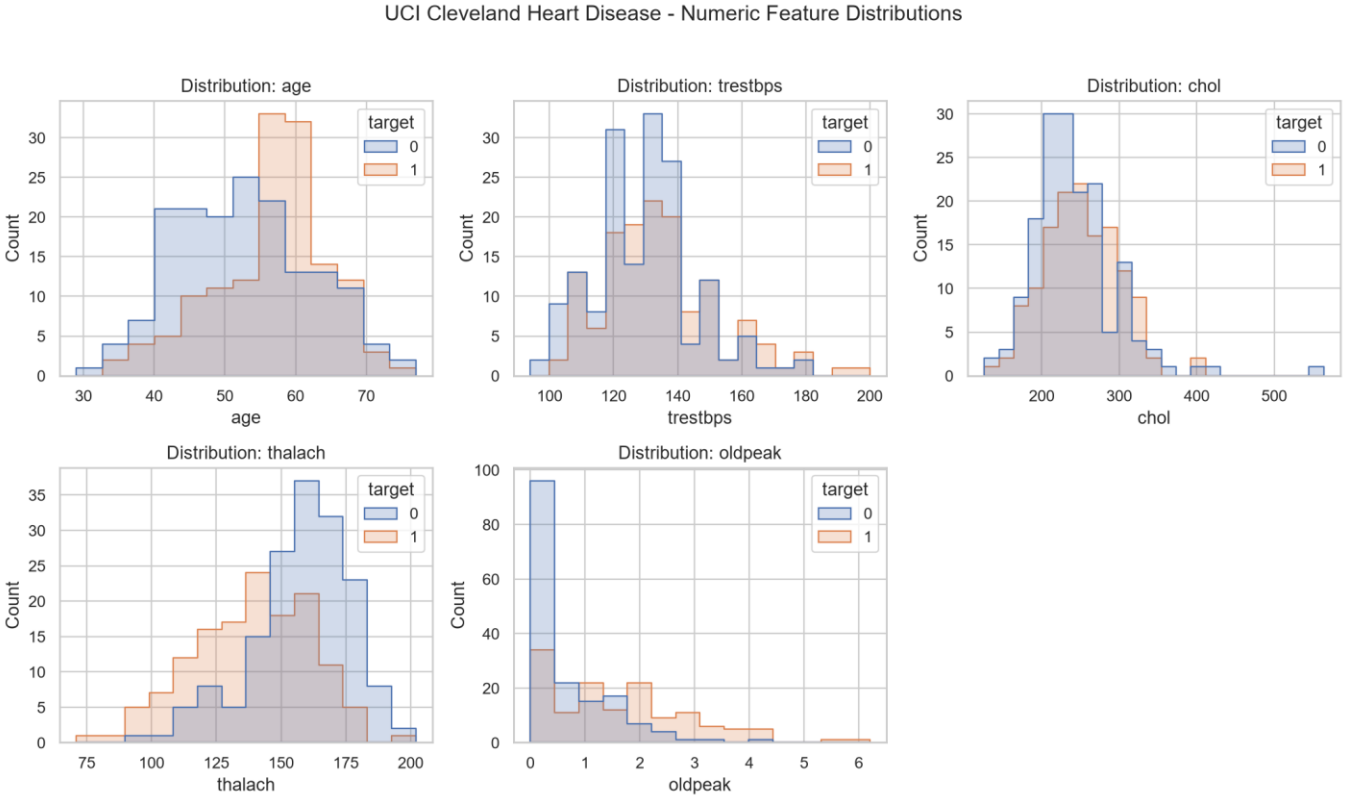
[1] UCI Machine Learning Repository, Heart Disease, https://archive.ics.uci.edu/dataset/45/heart+disease (dataset description, class definition, missing values and CC BY 4.0 attribution).

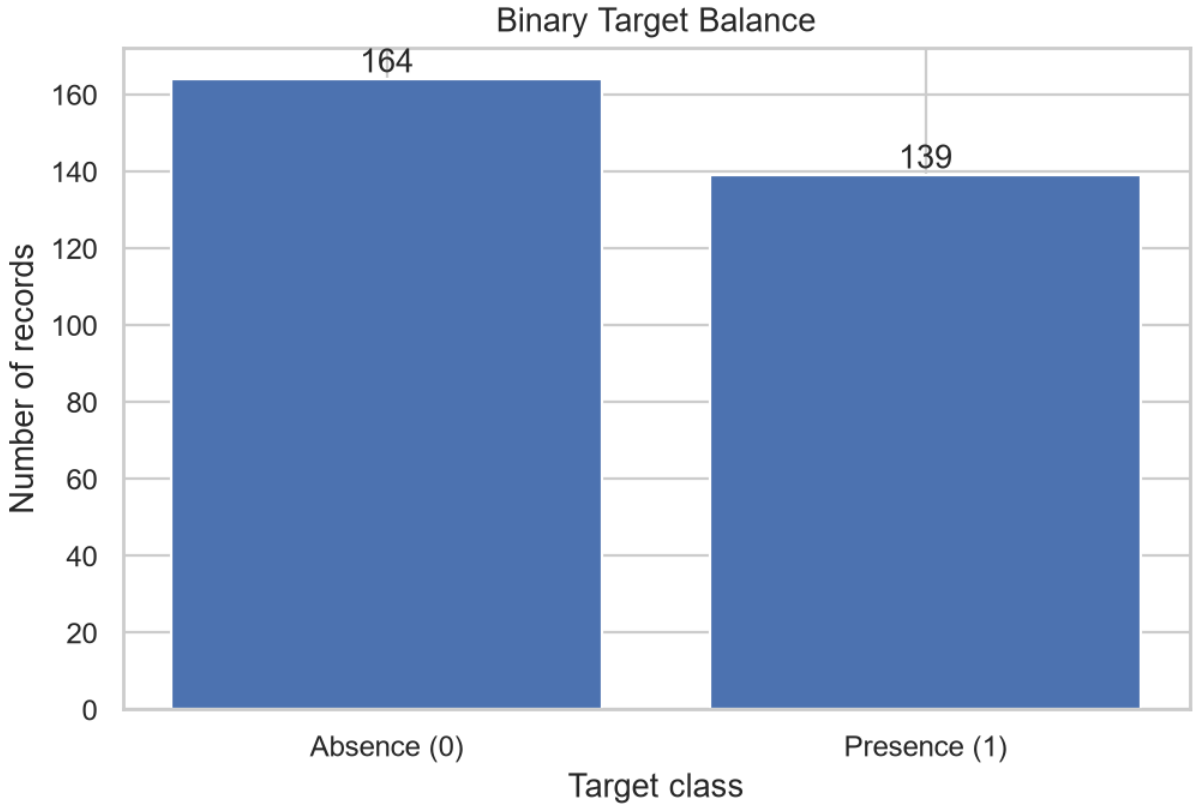
[2] MLflow tracking/autologging, https://mlflow.org/docs/latest/ml/tracking/autolog/ . [3] FastAPI testing, https://fastapi.tiangolo.com/tutorial/testing/ .

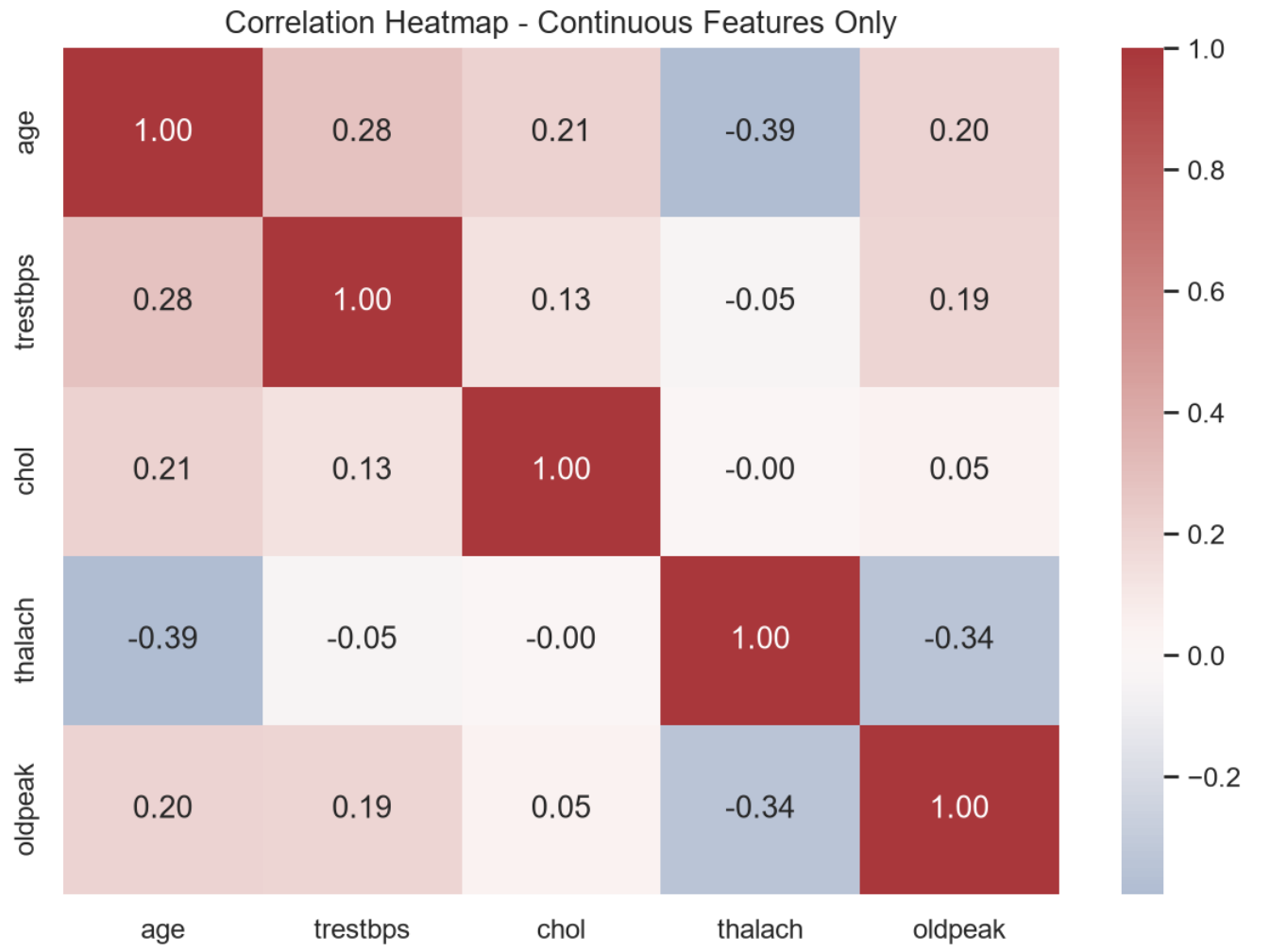
[4] Prometheus Python client, https://prometheus.github.io/client_python/ . [5] Kubernetes external LoadBalancer services, https://kubernetes.io/docs/tasks/access-application-cluster/create-external-load-balancer/ .

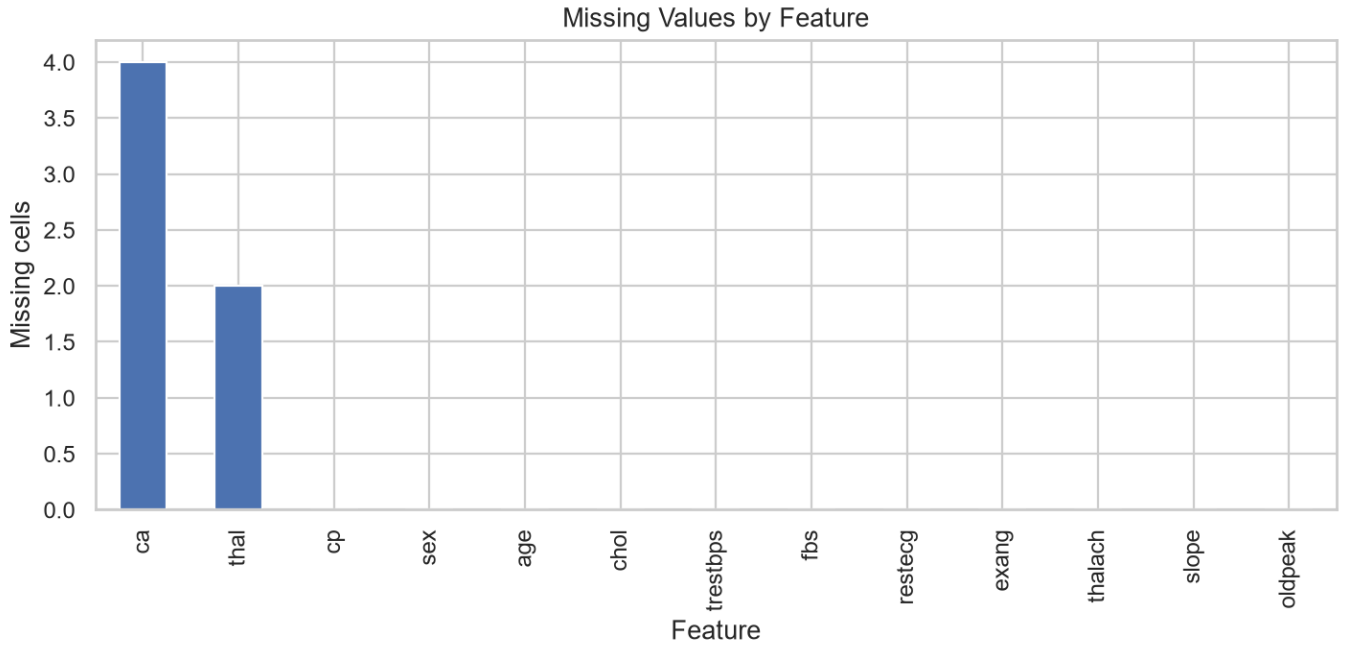
Final evidence checklist: add your GitHub repository URL, replace metrics only with values from the real run, save EDA/MLflow/Swagger/Docker/Grafana/GitHub Actions/Kubernetes screenshots as relevant, and include a short screen recording of the end-to-end flow.

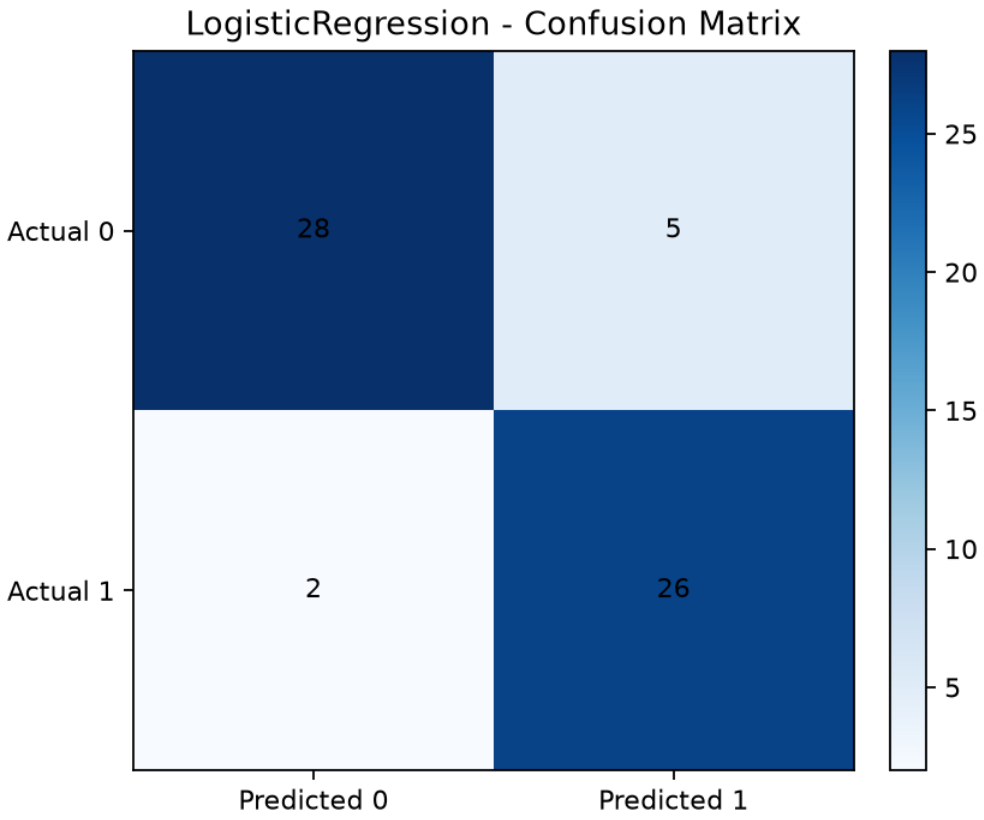
Reports/Figures:

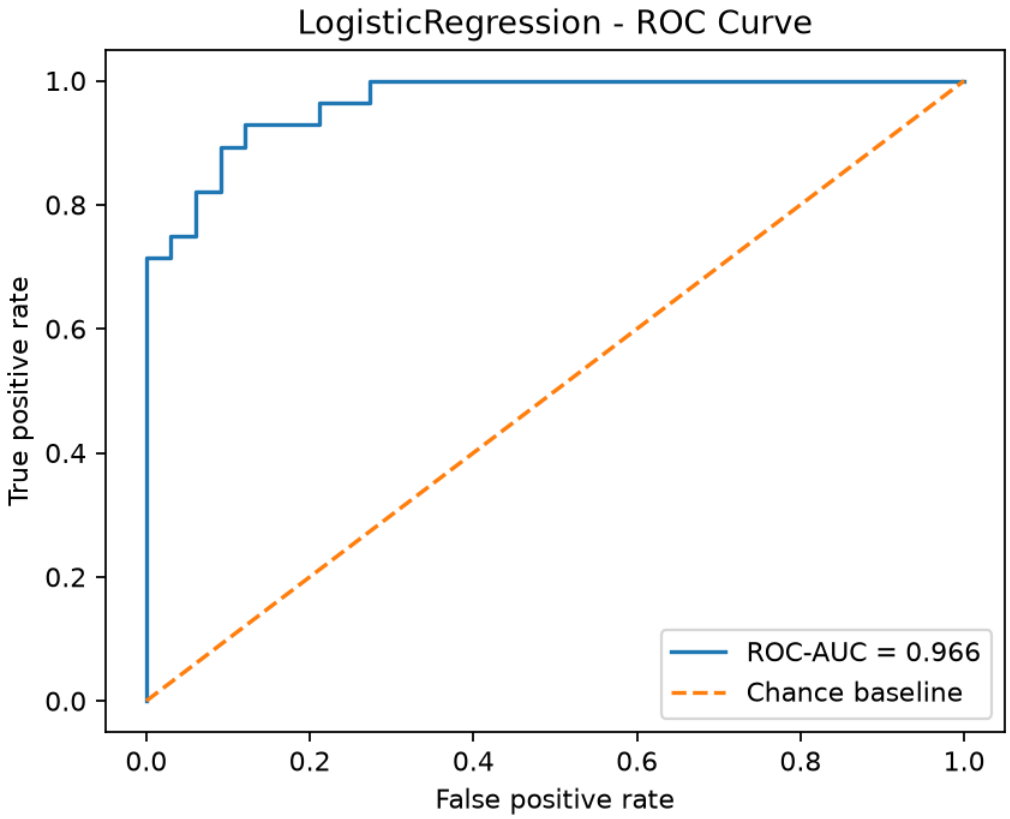

UCI Cleveland Heart Disease - Numeric Feature Distributions
Distribution: age
Distribution: trestbps
Distribution: chol
Distribution: thalach
Distribution: oldpeak
target
0
1
Count
age
trestbps
chol
thalach
oldpeak

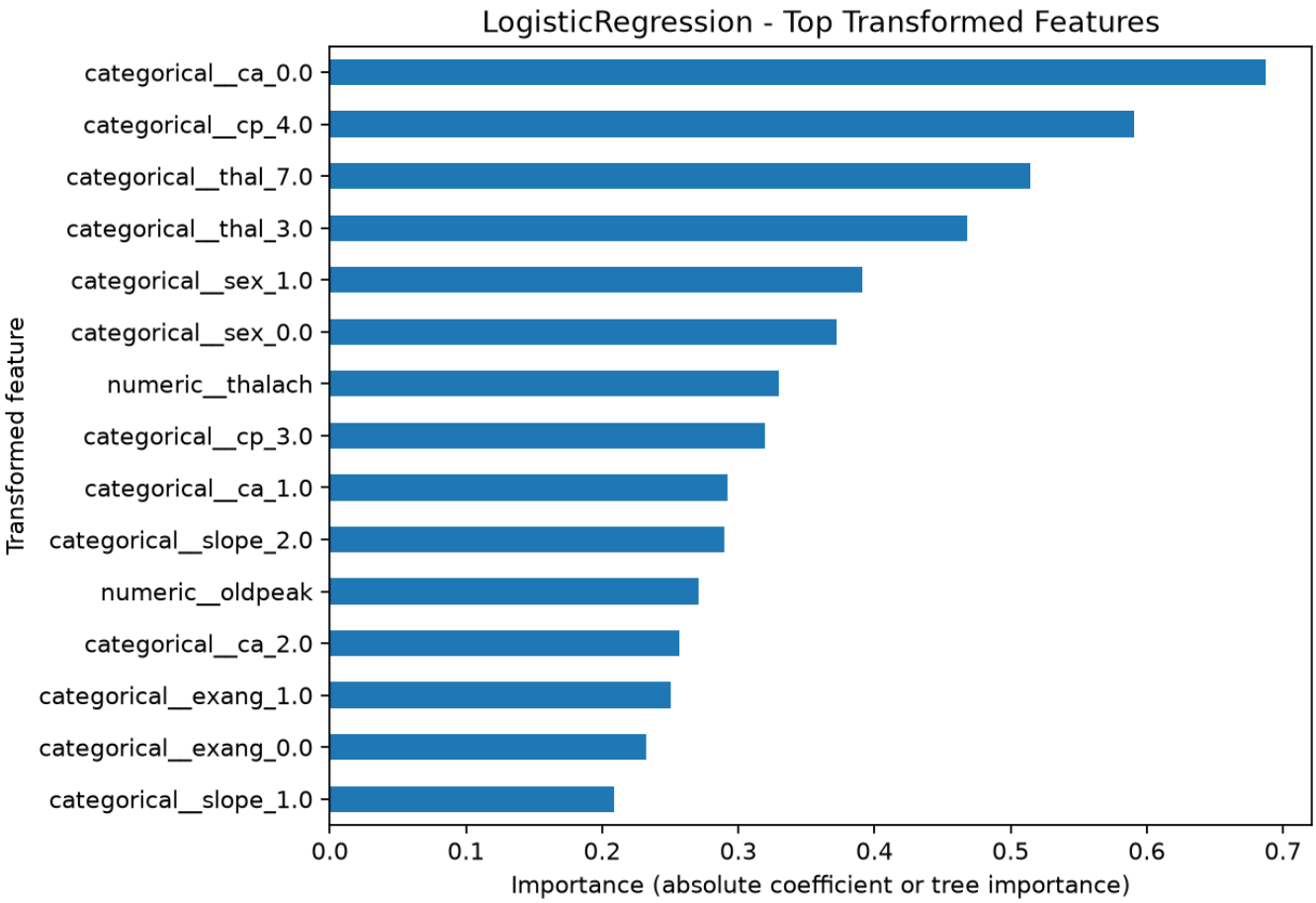

Binary Target Balance
164
139
Number of records
Absence (0)
Presence (1)
Target class

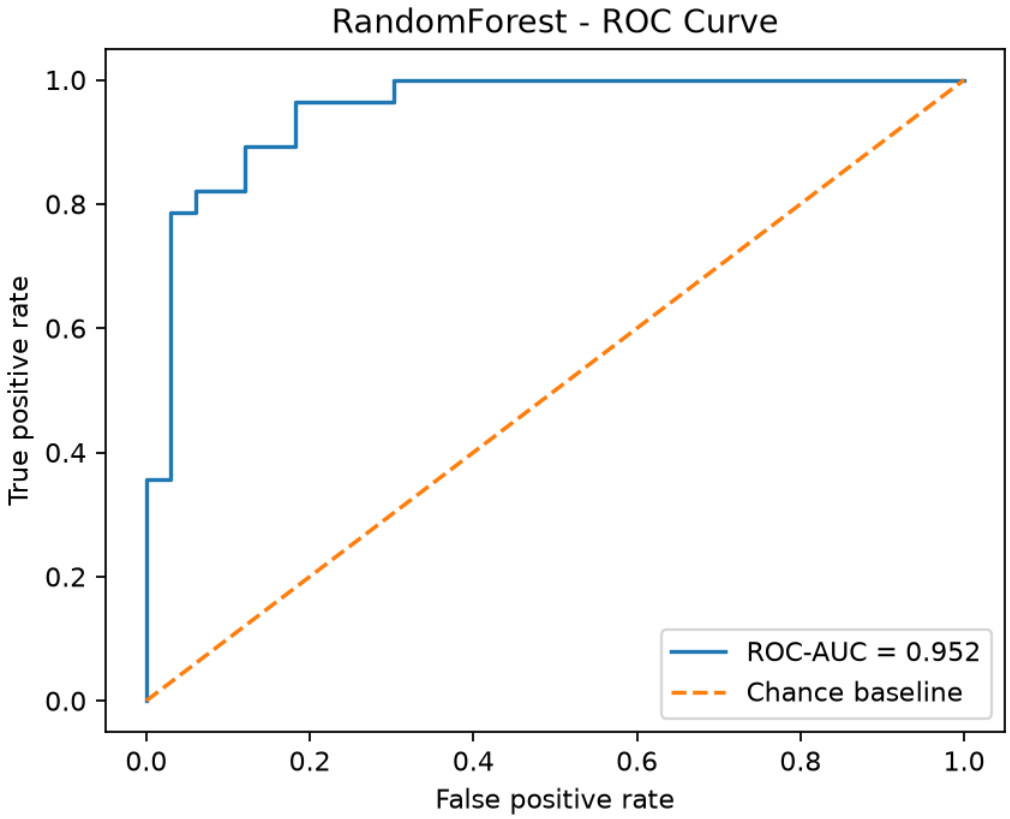

Correlation Heatmap - Continuous Features Only
age
trestbps
chol
thalach
oldpeak
1.00
0.28
0.21
-0.39
0.20
0.28
1.00
0.13
-0.05
0.19
0.21
0.13
1.00
-0.00
0.05
-0.39
-0.05
-0.00
1.00
-0.34
0.20
0.19
0.05
-0.34
1.00
1.0
0.8
0.6
0.4
0.2
0.0
−0.2

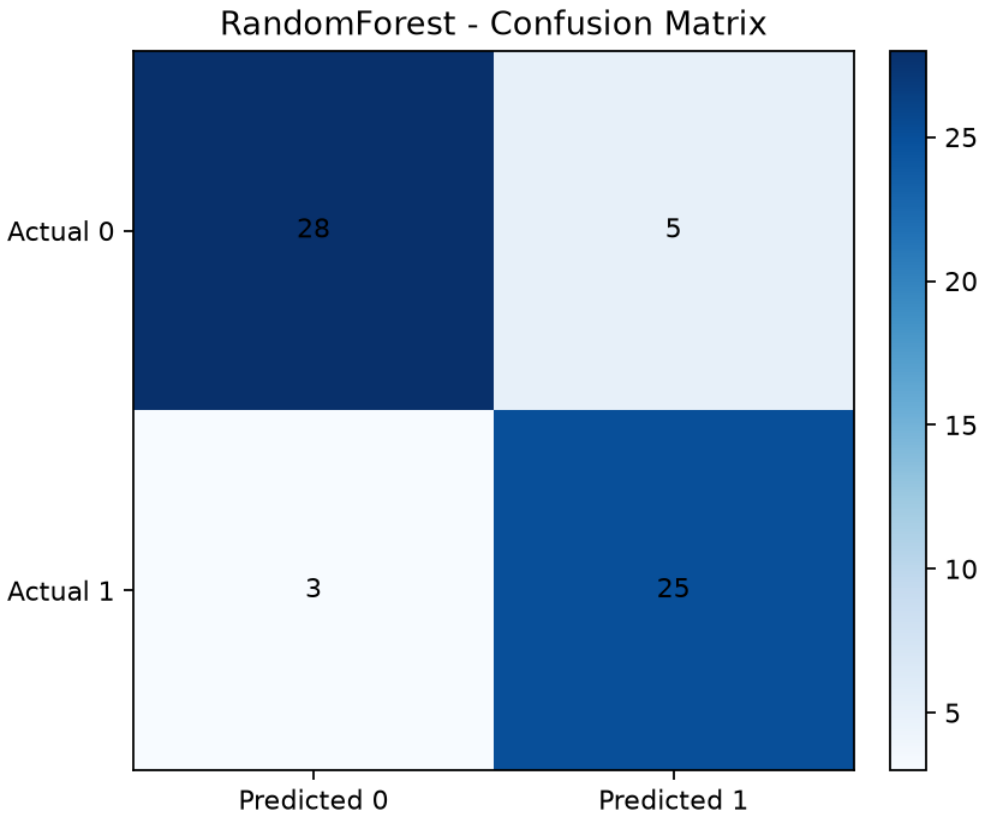

Missing Values by Feature
Missing cells
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
ca
thal
cp
sex
age
chol
trestbps
fbs
restecg
exang
thalach
slope
oldpeak
Feature

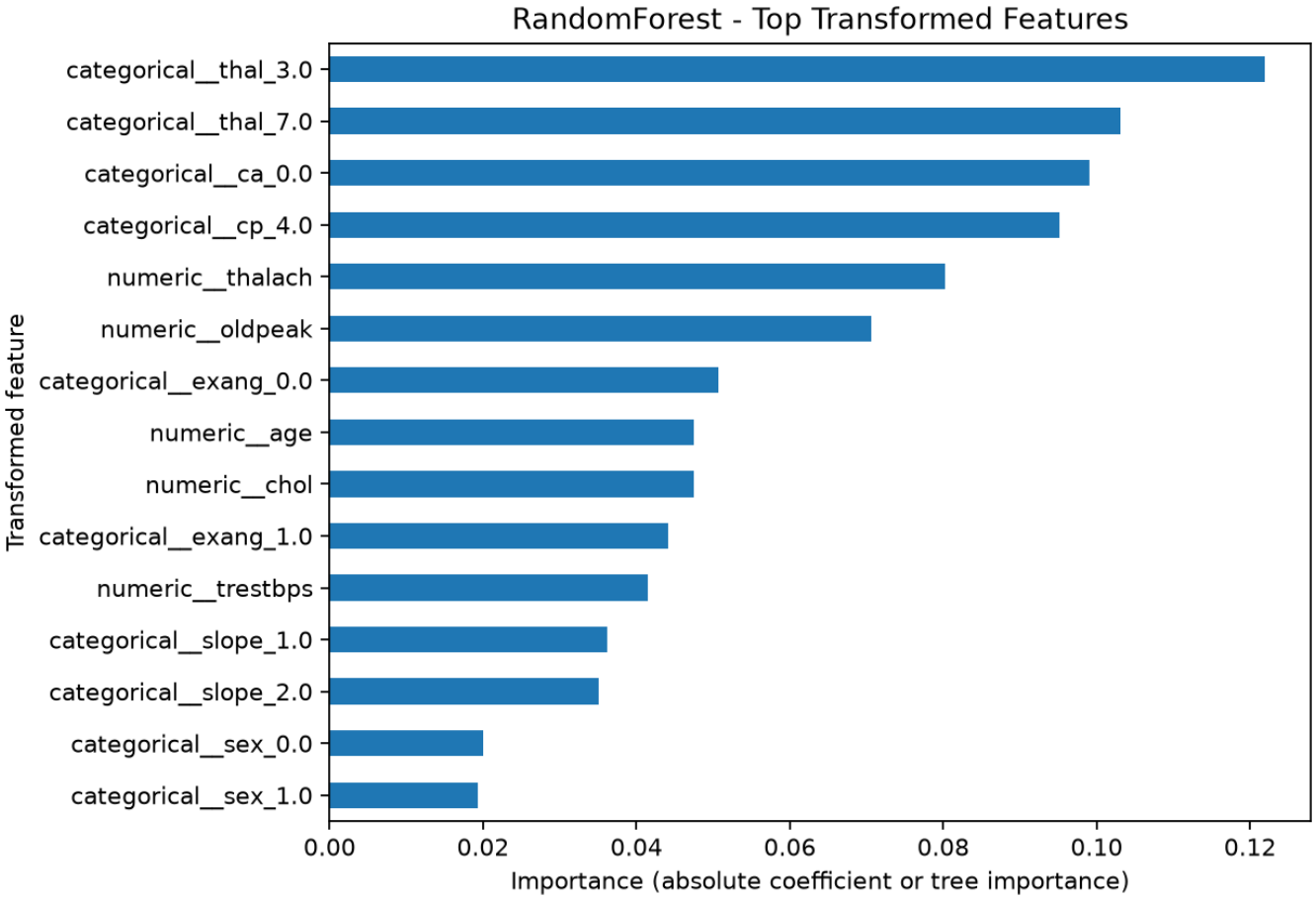

LogisticRegression - Confusion Matrix
Actual 0
Actual 1
28
5
2
26
Predicted 0
Predicted 1
25
20
15
10
5


LogisticRegression - ROC Curve
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC-AUC = 0.966
Chance baseline

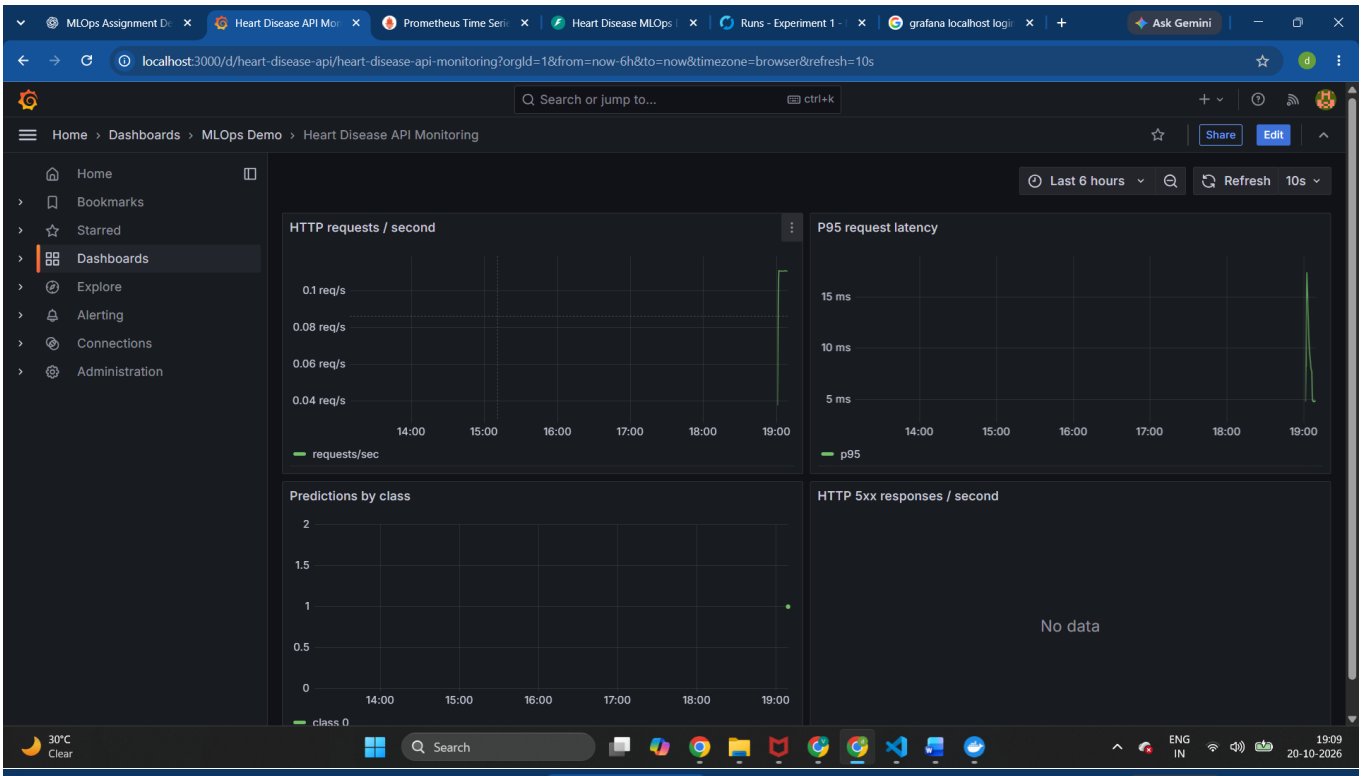

LogisticRegression - Top Transformed Features
categorical__ca_0.0
categorical__cp_4.0
categorical__thal_7.0
categorical__thal_3.0
categorical__sex_1.0
categorical__sex_0.0
numeric__thalach
categorical__cp_3.0
categorical__ca_1.0
categorical__slope_2.0
numeric__oldpeak
categorical__ca_2.0
categorical__exang_1.0
categorical__exang_0.0
categorical__slope_1.0
Transformed feature
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Importance (absolute coefficient or tree importance)

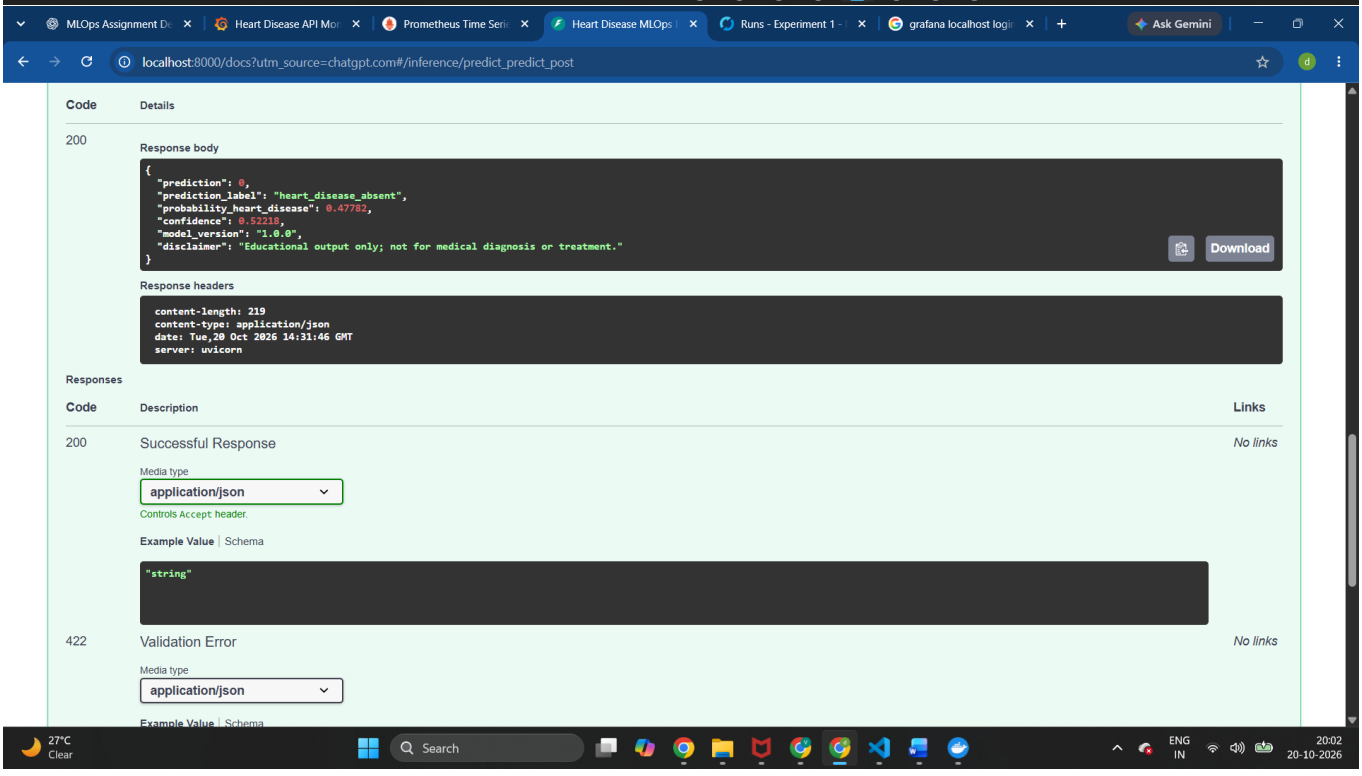

RandomForest - ROC Curve
True positive rate
False positive rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC-AUC = 0.952
Chance baseline

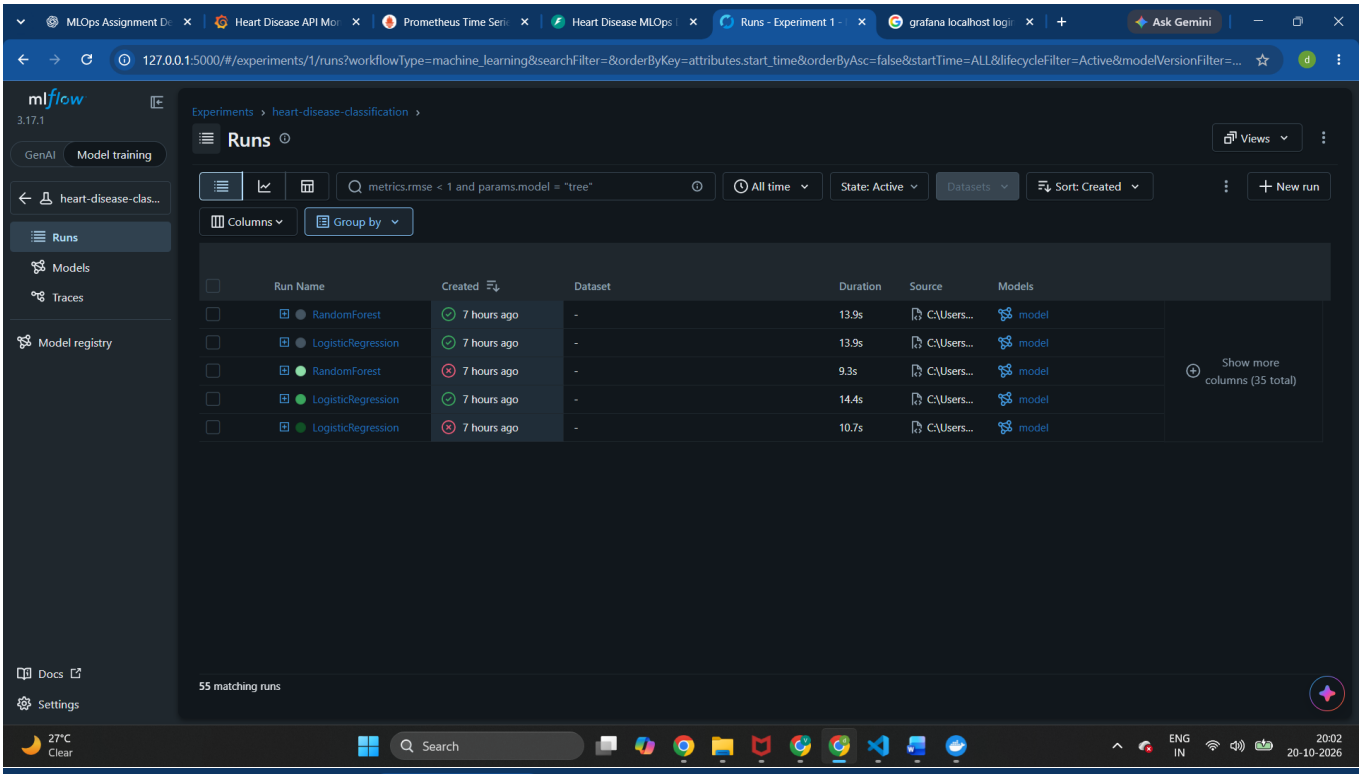

RandomForest - Confusion Matrix
Actual 0
Actual 1
28
5
3
25
Predicted 0
Predicted 1
25
20
15
10
5

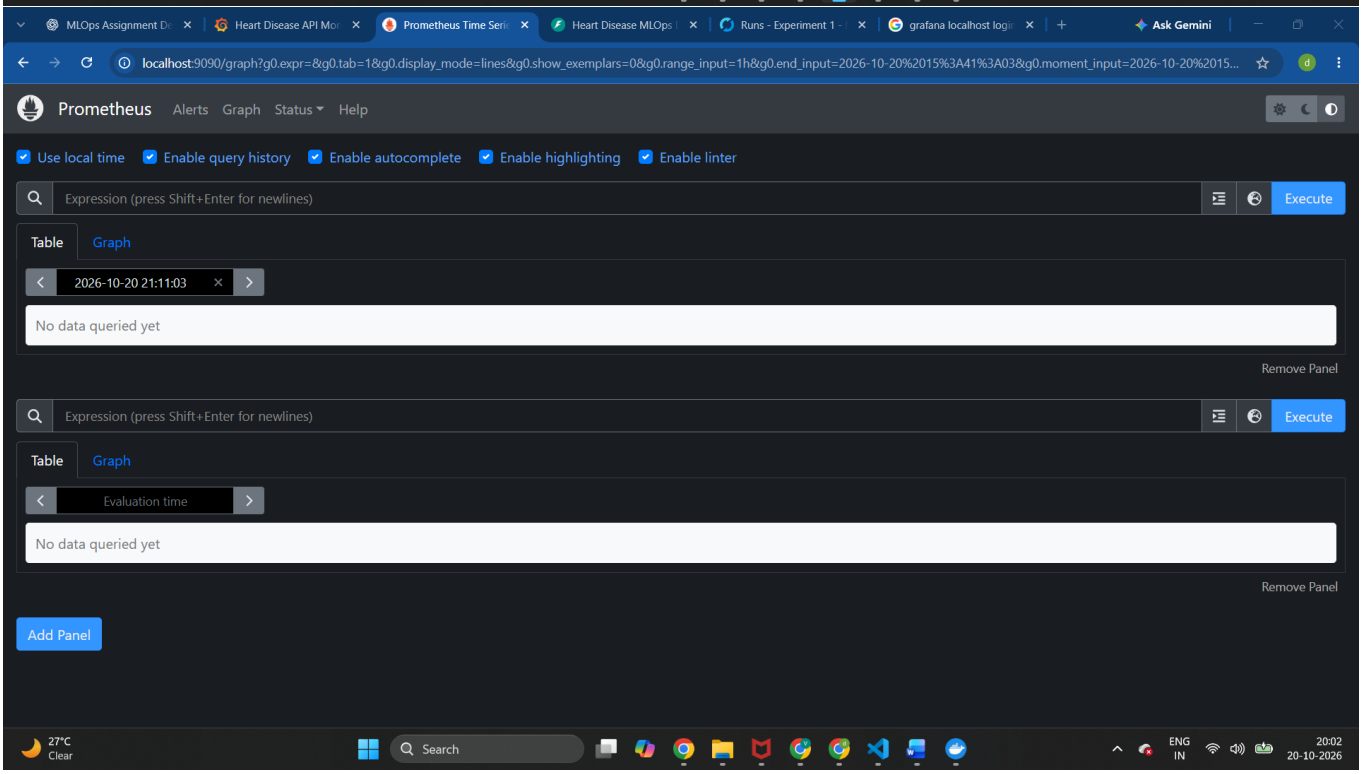

RandomForest - Top Transformed Features
Transformed feature
categorical__thal_3.0
categorical__thal_7.0
categorical__ca_0.0
categorical__cp_4.0
numeric__thalach
numeric__oldpeak
categorical__exang_0.0
numeric__age
numeric__chol
categorical__exang_1.0
numeric__trestbps
categorical__slope_1.0
categorical__slope_2.0
categorical__sex_0.0
categorical__sex_1.0
0.00
0.02
0.04
0.06
0.08
0.10
0.12
Importance (absolute coefficient or tree importance)

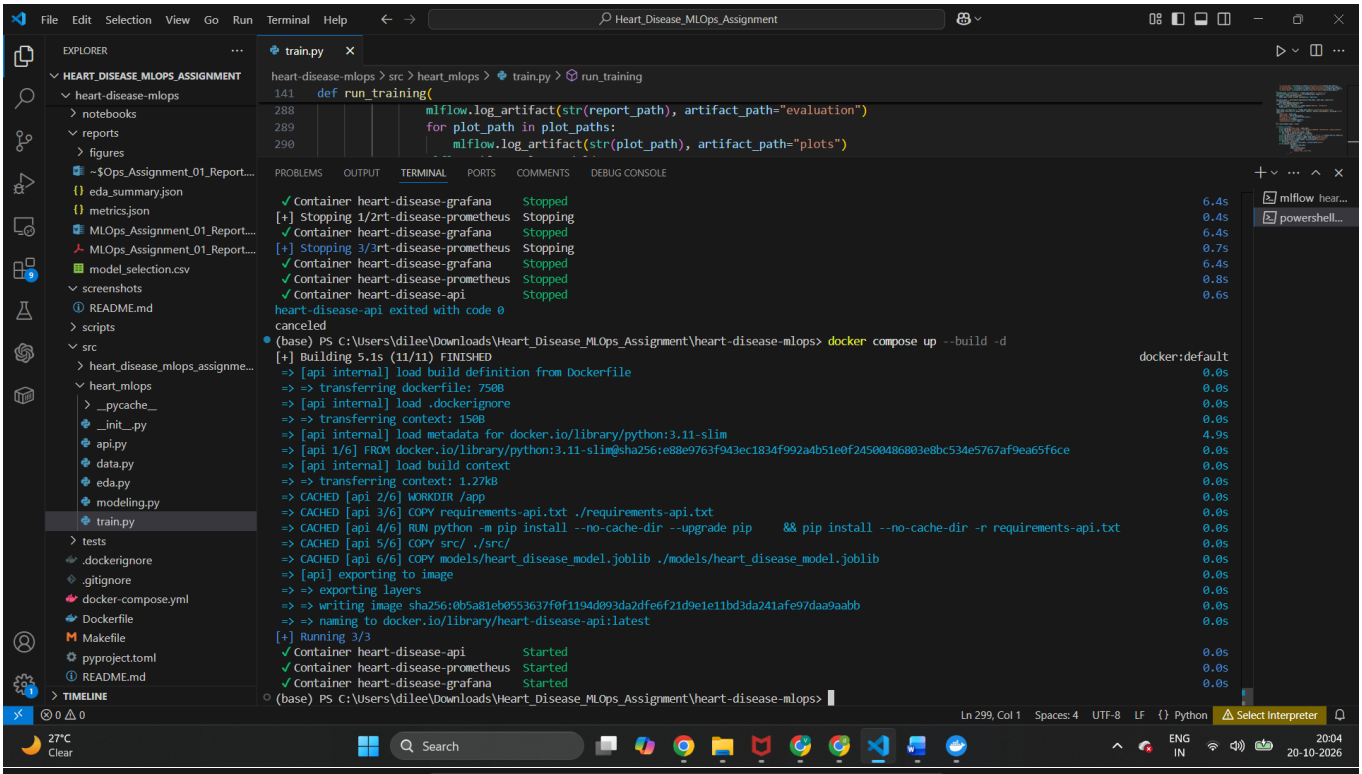
Screenshots:

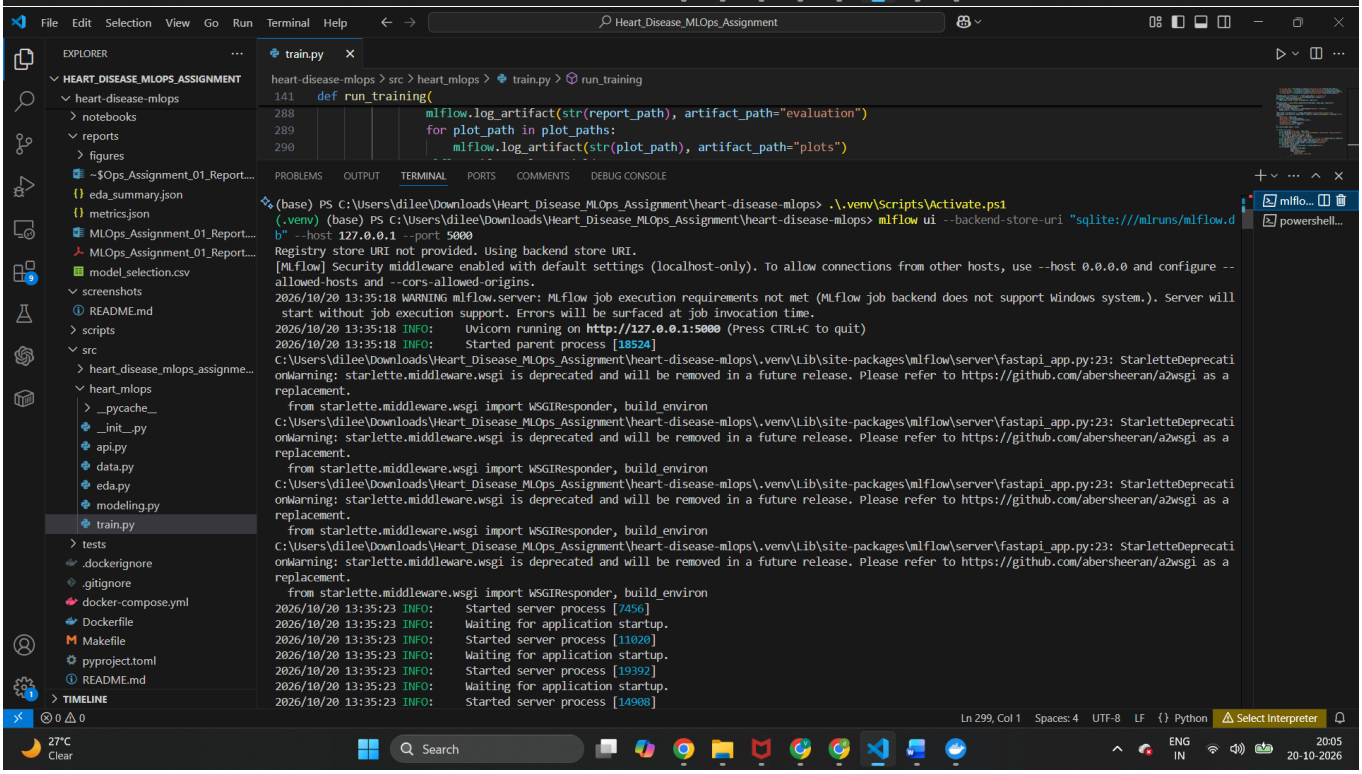





mlflow
3.17.1
GenAI
Model training
heart-disease-clas...
Runs
Models
Traces
Model registry
Docs
Settings
Experiments › heart-disease-classification ›
Runs
Views
metrics.rmse < 1 and params.model = "tree"
All time
State: Active
Datasets
Sort: Created
New run
Columns
Group by
Run Name
Created
Dataset
Duration
Source
Models
RandomForest 7 hours ago 13.9s C:\Users... model
LogisticRegression 7 hours ago 13.9s C:\Users... model
RandomForest 7 hours ago 9.3s C:\Users... model
LogisticRegression 7 hours ago 14.4s C:\Users... model
LogisticRegression 7 hours ago 10.7s C:\Users... model
Show more columns (35 total)
55 matching runs


Prometheus
Alerts
Graph
Status
Help
Use local time
Enable query history
Enable autocomplete
Enable highlighting
Enable linter
Expression (press Shift+Enter for newlines)
Execute
Table
Graph
2026-10-20 21:11:03
No data queried yet
Remove Panel
Evaluation time
Add Panel


File Edit Selection View Go Run Terminal Help
Heart_Disease_MLOps_Assignment
EXPLORER
HEART_DISEASE_MLOPS_ASSIGNMENT
heart-disease-mlops
notebooks
reports
figures
~$Ops_Assignment_01_Report....
eda_summary.json
metrics.json
MLOps_Assignment_01_Report....
MLOps_Assignment_01_Report....
model_selection.csv
screenshots
README.md
scripts
src
heart_disease_mlops_assignme...
heart_mlops
__pycache__
__init__.py
api.py
data.py
eda.py
modeling.py
train.py
tests
.dockerignore
.gitignore
docker-compose.yml
Dockerfile
Makefile
pyproject.toml
README.md
TIMELINE
train.py
heart-disease-mlops > src > heart_mlops > train.py > run_training
def run_training(
mlflow.log_artifact(str(report_path), artifact_path="evaluation")
for plot_path in plot_paths:
mlflow.log_artifact(str(plot_path), artifact_path="plots")
PROBLEMS OUTPUT TERMINAL PORTS COMMENTS DEBUG CONSOLE
✓ Container heart-disease-grafana Stopped 6.4s
[+] Stopping 1/2rt-disease-prometheus Stopping 0.4s
✓ Container heart-disease-grafana Stopped 6.4s
[+] Stopping 3/3rt-disease-prometheus Stopping 0.7s
✓ Container heart-disease-grafana Stopped 6.4s
✓ Container heart-disease-prometheus Stopped 0.8s
✓ Container heart-disease-api Stopped 0.6s
heart-disease-api exited with code 0
canceled
(base) PS C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops> docker compose up --build -d
[+] Building 5.1s (11/11) FINISHED docker:default
=> [api internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 750B 0.0s
=> [api internal] load .dockerignore 0.0s
=> => transferring context: 150B 0.0s
=> [api internal] load metadata for docker.io/library/python:3.11-slim 4.9s
=> [api 1/6] FROM docker.io/library/python:3.11-slim@sha256:e88e9763f943ec1834f992a4b51e0f24500486803e8bc534e5767af9ea65f6ce 0.0s
=> [api internal] load build context 0.0s
=> => transferring context: 1.27kB 0.0s
=> CACHED [api 2/6] WORKDIR /app 0.0s
=> CACHED [api 3/6] COPY requirements-api.txt ./requirements-api.txt 0.0s
=> CACHED [api 4/6] RUN python -m pip install --no-cache-dir --upgrade pip && pip install --no-cache-dir -r requirements-api.txt 0.0s
=> CACHED [api 5/6] COPY src/ ./src/ 0.0s
=> CACHED [api 6/6] COPY models/heart_disease_model.joblib ./models/heart_disease_model.joblib 0.0s
=> [api] exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:0b5a81eb0553637f0f1194d093da2dfe6f21d9e1e11bd3da241afe97daa9aabb 0.0s
=> => naming to docker.io/library/heart-disease-api:latest 0.0s
[+] Running 3/3
✓ Container heart-disease-api Started 0.0s
✓ Container heart-disease-prometheus Started 0.0s
✓ Container heart-disease-grafana Started 0.0s
(base) PS C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops>
mlflow hear...
powershell...
Ln 299, Col 1 Spaces: 4 UTF-8 LF {} Python Select Interpreter
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ENG IN
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TIMELINE
train.py
heart-disease-mlops > src > heart_mlops > train.py > run_training
def run_training(
mlflow.log_artifact(str(report_path), artifact_path="evaluation")
for plot_path in plot_paths:
mlflow.log_artifact(str(plot_path), artifact_path="plots")
PROBLEMS OUTPUT TERMINAL PORTS COMMENTS DEBUG CONSOLE
(base) PS C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops> .\.venv\Scripts\Activate.ps1
(.venv) (base) PS C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops> mlflow ui --backend-store-uri "sqlite:///mlruns/mlflow.db" --host 127.0.0.1 --port 5000
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
2026/10/20 13:35:18 WARNING mlflow.server: MLflow job execution requirements not met (MLflow job backend does not support Windows system.). Server will start without job execution support. Errors will be surfaced at job invocation time.
2026/10/20 13:35:18 INFO: Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
2026/10/20 13:35:18 INFO: Started parent process [18524]
C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops\.venv\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops\.venv\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops\.venv\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
C:\Users\dilee\Downloads\Heart_Disease_MLOps_Assignment\heart-disease-mlops\.venv\Lib\site-packages\mlflow\server\fastapi_app.py:23: StarletteDeprecationWarning: starlette.middleware.wsgi is deprecated and will be removed in a future release. Please refer to https://github.com/abersheeran/a2wsgi as a replacement.
from starlette.middleware.wsgi import WSGIResponder, build_environ
2026/10/20 13:35:23 INFO: Started server process [7456]
2026/10/20 13:35:23 INFO: Waiting for application startup.
2026/10/20 13:35:23 INFO: Started server process [11020]
2026/10/20 13:35:23 INFO: Waiting for application startup.
2026/10/20 13:35:23 INFO: Started server process [19392]
2026/10/20 13:35:23 INFO: Waiting for application startup.
2026/10/20 13:35:23 INFO: Started server process [14908]
mlflo...
powershell...
Ln 299, Col 1 Spaces: 4 UTF-8 LF {} Python Select Interpreter
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